The Æther-flow Interpretation of Relativity

Primitive-Reservoir Observer-Reduction
Observer-Relevant Pre-Core Exact-Shell Transport Core
from Exact-Shell-Transport-Core-Generating Substrate Sub-Pre-Core Dynamics

Abstract

The current primitive-reservoir observer-reduction frontier already spent the honest internal collapse available at full pre-core scope. The active transport-core theorem showed that the full exact-shell-core-generating substrate pre-core package contains benchmark-neutral ambient body directions and that the live burden factors instead through a smaller observer-relevant pre-core exact-shell transport core. After that step, another same-output deeper-origin relay that merely restates that same transport core is no longer the honest default move. The next admissible benchmark-facing gain must either derive or justify that transport core from still more primitive substrate structure, or identify a genuinely smaller theorem-level observer-relevant datum strictly inside it.

The present note takes the first route. For each sector it introduces *exact-shell-transport-core-generating substrate sub-pre-core dynamics*: a finite-dimensional sub-pre-core state space, a compact convex sub-pre-core body over the recorded transport-core body, a smooth sub-pre-core minimizer on that fixed body, and a closed exact sub-pre-core shell projecting diffeomorphically to the recorded pre-core exact-shell transport shell. The current observer-relevant pre-core exact-shell transport core is then recovered canonically as the projected exact-shell transport image of that deeper sub-pre-core package.

The benchmark boundary remains fixed throughout. The one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, the low-energy matter law is not enlarged, and no stronger promotion language is licensed. The positive result is therefore narrow but benchmark facing: the current transport core is no longer left as the primitive stopping point on the active line. The remaining honest burden moves one layer deeper again, to the exact-shell-transport-core-generating substrate sub-pre-core dynamics themselves or to a sharper collapse strictly inside that package.

1. Introduction

The active derivational program of Æther-flow Interpretation of Relativity has again reached a stage where the honest next move is constrained more sharply than "go one layer deeper." The current active-primary theorem already removed the full pre-core package as the live stopping point by proving that only a smaller observer-relevant pre-core exact-shell transport core remains benchmark-facingly live there. That result also fixed the admissible next alternatives: derive or justify that transport core from still more primitive substrate structure, or prove a genuinely smaller collapse strictly inside it.

The present note takes the default first route. Its role is not to reopen the full pre-core package whose discarded ambient directions have already been screened as benchmark neutral. Its role is

to show that even the now-live transport core can itself be presented as the projected exact-shell transport image of a deeper substrate package. In that sense the current transport core is no longer the primitive place where the active line has to stop.

Two features matter here. First, the theorem targets the transport core itself rather than returning to the already-screened larger pre-core package, so it respects the routing safeguard against recursive depth drift. Second, the deeper sub-pre-core package is not merely another renaming of the current transport core. It adds an explicit sub-pre-core state space, sub-pre-core body, sub-pre-core-to-transport-core projection, sub-pre-core minimizer, and deeper exact-shell transport below the current frontier. The current transport core is therefore recovered from more primitive substrate data rather than simply reasserted.

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

The present note stays strictly inside that boundary. It does not alter the benchmark exact-closure package. It does not introduce a second operative metric or a new low-energy matter law. It only proves that the current observer-relevant pre-core exact-shell transport core can itself be generated from more primitive sub-pre-core substrate dynamics.

2. Fixed Inputs

Definition 1 (Recorded primitive continuation data)**.** *Fix the following continuation data from the active primitive-reservoir observer-reduction line.*

1. *For each sector label* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*, a primitive forcing shell*

$$\mathcal{I}_\sigma^{\mathrm{for}},$$

a visible reduction map

$$\mathcal{R}_\sigma : \mathcal{I}_\sigma^{\mathrm{for}} \to X_\sigma,$$

and shell-level current and equilibrium carriers

$$\mathfrak{C}_\sigma^{\mathrm{cur}} : \mathcal{I}_\sigma^{\mathrm{for}} \to Z_\sigma^{\mathrm{cur}}, \qquad \mathfrak{C}_\sigma^{\mathrm{eq}} : \mathcal{I}_\sigma^{\mathrm{for}} \to Z_\sigma^{\mathrm{eq}}.$$

2. *The product shell*

$$\mathcal{S}_{\mathrm{prim}} \equiv \mathcal{I}_{\mathrm{car}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{cur}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{hom}}^{\mathrm{for}}.$$

3. *The recorded metric and ordinary-matter outputs*

$$\mathscr{G}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \to \{\textit{observer metrics}\}, \qquad \mathscr{T}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \times \Psi \to \{\textit{observer stress tensors}\},$$

satisfying

$$\mathscr{G}_{\mathrm{out}}(\mathbf{w}) = g^{\sharp}, \qquad \mathscr{T}_{\mathrm{out}}(\mathbf{w}, \psi) = T^{\mathrm{matter}}_{\mu\nu}[g^{\sharp}, \psi] \tag{1}$$

for every $\mathbf{w} \in \mathcal{S}_{\mathrm{prim}}$ *and every matter configuration* $\psi \in \Psi$.

Definition 2 (Recorded proto-germ exact-shell target)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$. *The active line already fixes:*

1. *the same substrate target and pairing*

$$E^{\mathrm{sub}}_{\sigma}, \qquad \mathfrak{s}^{\mathrm{sub}}_{\sigma} : E^{\mathrm{sub}}_{\sigma} \times E^{\mathrm{sub}}_{\sigma} \to \mathbb{R};$$

2. *a hidden fiber space* $\mathcal{H}^{\mathrm{hid}}_{\sigma}$ *with compact hidden body* $D^{\mathrm{hid}}_{\sigma} \subset \mathcal{H}^{\mathrm{hid}}_{\sigma}$*;*

3. *a finite-dimensional proto-germ state space* $\mathcal{U}^{\mathrm{proto}}_{\sigma}$*, a compact convex proto-germ body* $\mathcal{B}^{\mathrm{proto}}_{\sigma} \subset \mathcal{U}^{\mathrm{proto}}_{\sigma}$*, a smooth proto-germ restriction section*

$$\Gamma^{\mathrm{proto}}_{\sigma} : \mathcal{U}^{\mathrm{proto}}_{\sigma} \to E^{\mathrm{sub}}_{\sigma},$$

and the exact proto-germ shell

$$\mathcal{Z}^{\mathrm{proto}}_{\sigma} \equiv \left(\Gamma^{\mathrm{proto}}_{\sigma}\right)^{-1}(0) \cap \mathcal{B}^{\mathrm{proto}}_{\sigma};$$

4. *a closed embedded proto zero core* $\mathcal{Z}^{0,\mathrm{proto}}_{\sigma} \subset \mathcal{Z}^{\mathrm{proto}}_{\sigma}$ *and a split diffeomorphism*

$$\Phi^{\mathrm{proto}}_{\sigma} : \mathcal{Z}^{0,\mathrm{proto}}_{\sigma} \times D^{\mathrm{hid}}_{\sigma} \xrightarrow{\cong} \mathcal{Z}^{\mathrm{proto}}_{\sigma};$$

5. *a hidden charge*

$$\mathcal{Q}^{\mathrm{proto}}_{\sigma} : \mathcal{H}^{\mathrm{hid}}_{\sigma} \to Q^{\mathrm{eqn}}_{\sigma},$$

a smooth observer-compatible exact proto-germ zero-response realization

$$\zeta^{\mathrm{proto}}_{\sigma} : \mathcal{I}^{\mathrm{for}}_{\sigma} \hookrightarrow \mathcal{Z}^{0,\mathrm{proto}}_{\sigma},$$

and a fixed-data solvability / coercive package on $\mathcal{Z}^{0,\mathrm{proto}}_{\sigma}$*, recorded through* $\widetilde{K}^{0,\mathrm{proto}}_{\sigma}$ *and* $P^{\mathrm{proto,co}}_{\sigma}$.

Definition 3 (Recorded current transport nucleus)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$. *The recorded current transport nucleus* $\mathfrak{Q}^{\mathrm{nuc}}$ *for sector* σ *consists of:*

1. *the pre-proto-germ state space* $\mathcal{V}^{\mathrm{preproto}}_{\sigma}$*, compact convex body* $\mathcal{B}^{\mathrm{preproto}}_{\sigma} \subset \mathcal{V}^{\mathrm{preproto}}_{\sigma}$*, projection*

$$\Pi^{\mathrm{preproto}}_{\sigma} : \mathcal{V}^{\mathrm{preproto}}_{\sigma} \to \mathcal{U}^{\mathrm{proto}}_{\sigma},$$

and proto section

$$\mathfrak{f}^{\mathrm{preproto,proto}}_{\sigma} : \mathcal{B}^{\mathrm{proto}}_{\sigma} \to \mathcal{B}^{\mathrm{preproto}}_{\sigma}$$

satisfying

$$\Pi_\sigma^{\mathrm{preproto}}(\mathcal{B}_\sigma^{\mathrm{preproto}}) = \mathcal{B}_\sigma^{\mathrm{proto}}, \qquad \Pi_\sigma^{\mathrm{preproto}} \circ \mathfrak{f}_\sigma^{\mathrm{preproto,proto}} = \mathrm{id}_{\mathcal{B}_\sigma^{\mathrm{proto}}};$$

2. *the restriction section*

$$\Gamma_\sigma^{\mathrm{preproto}} : \mathcal{V}_\sigma^{\mathrm{preproto}} \to E_\sigma^{\mathrm{sub}},$$

the induced functional

$$\mathscr{W}_\sigma^{\mathrm{preproto}}(q) \equiv \mathfrak{s}_\sigma^{\mathrm{sub}}(\Gamma_\sigma^{\mathrm{preproto}}(q), \Gamma_\sigma^{\mathrm{preproto}}(q)),$$

and the exact shell

$$\mathcal{Z}_\sigma^{\mathrm{preproto}} \equiv (\Gamma_\sigma^{\mathrm{preproto}})^{-1}(0) \cap \mathcal{B}_\sigma^{\mathrm{preproto}},$$

with

$$\Gamma_\sigma^{\mathrm{preproto}} \circ \mathfrak{f}_\sigma^{\mathrm{preproto,proto}} = \Gamma_\sigma^{\mathrm{proto}}, \qquad \Pi_\sigma^{\mathrm{preproto}}\big|_{\mathcal{Z}_\sigma^{\mathrm{preproto}}} : \mathcal{Z}_\sigma^{\mathrm{preproto}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{proto}};$$

3. *benchmark faithfulness, meaning preservation of the benchmark outputs* (1)*, no second operative metric, no enlarged low-energy matter law, and no premature promotion from adoption to completed derivation.*

Definition 4 (Recorded observer-relevant pre-nucleus exact-shell core)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*. The observer-relevant pre-nucleus exact-shell core* $\mathfrak{R}^{\mathrm{core}}$ *for sector* σ *consists of:*

1. *the pre-nucleus state space* $\mathcal{W}_\sigma^{\mathrm{prenuc}}$*, compact convex body* $\mathcal{B}_\sigma^{\mathrm{prenuc}} \subset \mathcal{W}_\sigma^{\mathrm{prenuc}}$*, affine surjection*

$$\Pi_\sigma^{\mathrm{prenuc}} : \mathcal{W}_\sigma^{\mathrm{prenuc}} \to \mathcal{V}_\sigma^{\mathrm{preproto}},$$

and fiber minimizer

$$\mathfrak{f}_\sigma^{\mathrm{prenuc,nuc}} : \mathcal{B}_\sigma^{\mathrm{preproto}} \to \mathcal{B}_\sigma^{\mathrm{prenuc}}$$

satisfying

$$\Pi_\sigma^{\mathrm{prenuc}}(\mathcal{B}_\sigma^{\mathrm{prenuc}}) = \mathcal{B}_\sigma^{\mathrm{preproto}}, \qquad \Pi_\sigma^{\mathrm{prenuc}} \circ \mathfrak{f}_\sigma^{\mathrm{prenuc,nuc}} = \mathrm{id}_{\mathcal{B}_\sigma^{\mathrm{preproto}}};$$

2. *a closed embedded exact pre-nucleus shell*

$$\mathcal{Z}_\sigma^{\mathrm{prenuc}} \subset \mathcal{B}_\sigma^{\mathrm{prenuc}}$$

satisfying

$$\mathfrak{f}_\sigma^{\mathrm{prenuc,nuc}}(\mathcal{Z}_\sigma^{\mathrm{preproto}}) \subset \mathcal{Z}_\sigma^{\mathrm{prenuc}}, \qquad \Pi_\sigma^{\mathrm{prenuc}}\big|_{\mathcal{Z}_\sigma^{\mathrm{prenuc}}} : \mathcal{Z}_\sigma^{\mathrm{prenuc}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{preproto}};$$

3. *benchmark faithfulness in the sense of Definition 3.*

Definition 5 (Recorded pre-core exact-shell transport core)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*. The recorded observer-relevant pre-core exact-shell transport core* $\mathfrak{S}^{\mathrm{tr}}$ *for sector* σ *consists of:*

1. *a compact transport-core body*

$$\mathcal{B}_\sigma^{\mathrm{precore,tr}},$$

an affine projection

$$\widehat{\Pi}_\sigma^{\mathrm{precore}} : \mathcal{B}_\sigma^{\mathrm{precore,tr}} \to \mathcal{B}_\sigma^{\mathrm{prenuc}},$$

and a smooth transport-core lift

$$\mathfrak{f}_\sigma^{\mathrm{precore,tr}} : \mathcal{B}_\sigma^{\mathrm{prenuc}} \to \mathcal{B}_\sigma^{\mathrm{precore,tr}}$$

satisfying

$$\widehat{\Pi}_\sigma^{\mathrm{precore}}(\mathcal{B}_\sigma^{\mathrm{precore,tr}}) = \mathcal{B}_\sigma^{\mathrm{prenuc}}, \qquad \widehat{\Pi}_\sigma^{\mathrm{precore}} \circ \mathfrak{f}_\sigma^{\mathrm{precore,tr}} = \mathrm{id}_{\mathcal{B}_\sigma^{\mathrm{prenuc}}};$$

2. *a closed embedded exact transport shell*

$$\mathcal{Z}_\sigma^{\mathrm{precore,tr}} \subset \mathcal{B}_\sigma^{\mathrm{precore,tr}}$$

satisfying

$$\mathfrak{f}_\sigma^{\mathrm{precore,tr}}(\mathcal{Z}_\sigma^{\mathrm{prenuc}}) \subset \mathcal{Z}_\sigma^{\mathrm{precore,tr}}, \qquad \widehat{\Pi}_\sigma^{\mathrm{precore}}\big|_{\mathcal{Z}_\sigma^{\mathrm{precore,tr}}} : \mathcal{Z}_\sigma^{\mathrm{precore,tr}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{prenuc}};$$

3. *benchmark faithfulness in the sense of Definition 4.*

Remark 1. *Definitions 1–5 are fixed inputs. The one operative metric, the ordinary matter route, the fixed proto-germ exact-shell target, the current transport nucleus, the recorded pre-nucleus exact-shell core, and the current observer-relevant pre-core exact-shell transport core are not reopened here. The present note asks only whether that current transport core can itself be generated from more primitive substrate dynamics.*

3. Exact-Shell-Transport-Core-Generating Substrate Sub-Pre-Core Dynamics

Definition 6 (Exact-shell-transport-core-generating substrate sub-pre-core dynamics)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*. An* exact-shell-transport-core-generating substrate sub-pre-core dynamics package $\mathfrak{U}^{\mathrm{subpre}}$ *for sector* σ *consists of the following data.*

1. *A finite-dimensional Euclidean sub-pre-core state space* $\mathcal{L}_\sigma^{\mathrm{subpre}}$*, a compact convex sub-pre-core body*

$$\mathcal{B}_\sigma^{\mathrm{subpre}} \subset \mathcal{L}_\sigma^{\mathrm{subpre}}$$

with nonempty interior, an affine surjection

$$\Pi_\sigma^{\mathrm{subpre}} : \mathcal{L}_\sigma^{\mathrm{subpre}} \to \mathcal{B}_\sigma^{\mathrm{precore,tr}},$$

and a smooth sub-pre-core minimizer

$$\mathfrak{f}_\sigma^{\mathrm{subpre,tr}} : \mathcal{B}_\sigma^{\mathrm{precore,tr}} \to \mathcal{B}_\sigma^{\mathrm{subpre}}$$

satisfying

$$\Pi_\sigma^{\mathrm{subpre}}(\mathcal{B}_\sigma^{\mathrm{subpre}}) = \mathcal{B}_\sigma^{\mathrm{precore,tr}}, \qquad \Pi_\sigma^{\mathrm{subpre}} \circ \mathfrak{f}_\sigma^{\mathrm{subpre,tr}} = \mathrm{id}_{\mathcal{B}_\sigma^{\mathrm{precore,tr}}}.$$

2. *A closed embedded exact sub-pre-core shell*

$$\mathcal{Z}_\sigma^{\mathrm{subpre}} \subset \mathcal{B}_\sigma^{\mathrm{subpre}}$$

satisfying

$$\mathfrak{f}_\sigma^{\mathrm{subpre,tr}}(\mathcal{Z}_\sigma^{\mathrm{precore,tr}}) \subset \mathcal{Z}_\sigma^{\mathrm{subpre}}, \qquad \Pi_\sigma^{\mathrm{subpre}}\big|_{\mathcal{Z}_\sigma^{\mathrm{subpre}}} : \mathcal{Z}_\sigma^{\mathrm{subpre}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{precore,tr}}.$$

3. *Benchmark faithfulness, meaning preservation of the benchmark outputs* (1)*, no second operative metric, no enlarged low-energy matter law, and no premature promotion from adoption to completed derivation.*

Remark 2. *Definition 6 is intentionally written over the recorded transport-core target of Definition 5. It does not reintroduce the already-screened larger pre-core package as live data. Its positive role is narrower: provide a more primitive substrate package whose projected body transport and exact-shell transport already generate the current transport core.*

4. Canonical Recovery of the Current Transport Core

Proposition 1 (Sub-pre-core dynamics generate the recorded transport core)**.** *Every benchmark-faithful sub-pre-core package in the sense of Definition 6 generates the recorded observer-relevant pre-core exact-shell transport core of Definition 5 by projected body transport and exact-shell transport.*

Proof. Definition 6(1) gives a compact convex sub-pre-core body together with an affine surjection

$$\Pi_\sigma^{\mathrm{subpre}} : \mathcal{L}_\sigma^{\mathrm{subpre}} \to \mathcal{B}_\sigma^{\mathrm{precore,tr}}$$

and a smooth minimizer

$$\mathfrak{f}_\sigma^{\mathrm{subpre,tr}} : \mathcal{B}_\sigma^{\mathrm{precore,tr}} \to \mathcal{B}_\sigma^{\mathrm{subpre}}$$

satisfying

$$\Pi_\sigma^{\mathrm{subpre}}(\mathcal{B}_\sigma^{\mathrm{subpre}}) = \mathcal{B}_\sigma^{\mathrm{precore,tr}}, \qquad \Pi_\sigma^{\mathrm{subpre}} \circ \mathfrak{f}_\sigma^{\mathrm{subpre,tr}} = \mathrm{id}_{\mathcal{B}_\sigma^{\mathrm{precore,tr}}}.$$

Definition 6(2) gives a closed embedded exact sub-pre-core shell projecting diffeomorphically to the recorded transport shell:

$$\Pi_\sigma^{\mathrm{subpre}}\big|_{\mathcal{Z}_\sigma^{\mathrm{subpre}}} : \mathcal{Z}_\sigma^{\mathrm{subpre}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{precore,tr}}.$$

Therefore the recorded body transport and exact-shell transport of Definition 5 arise as the projected exact-shell image of the deeper sub-pre-core package. Benchmark faithfulness is preserved by Definition 6(3). □

Proposition 2 (Downstream transport to the recorded pre-nucleus core persists)**.** *Fix a benchmark-faithful sub-pre-core package* $\mathfrak{U}^{\mathrm{subpre}}$*. Then*

$$(\widehat{\Pi}_\sigma^{\mathrm{precore}} \circ \Pi_\sigma^{\mathrm{subpre}})(\mathcal{B}_\sigma^{\mathrm{subpre}}) = \mathcal{B}_\sigma^{\mathrm{prenuc}},$$

$$(\widehat{\Pi}_\sigma^{\mathrm{precore}} \circ \Pi_\sigma^{\mathrm{subpre}}) \circ \mathfrak{f}_\sigma^{\mathrm{subpre,tr}} \circ \mathfrak{f}_\sigma^{\mathrm{precore,tr}} = \mathrm{id}_{\mathcal{B}_\sigma^{\mathrm{prenuc}}},$$

and the composite shell transport

$$(\widehat{\Pi}_\sigma^{\mathrm{precore}} \circ \Pi_\sigma^{\mathrm{subpre}})\big|_{\mathcal{Z}_\sigma^{\mathrm{subpre}}} : \mathcal{Z}_\sigma^{\mathrm{subpre}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{prenuc}}$$

is a diffeomorphism. Consequently

$$(\Pi_\sigma^{\mathrm{prenuc}} \circ \widehat{\Pi}_\sigma^{\mathrm{precore}} \circ \Pi_\sigma^{\mathrm{subpre}})\big|_{\mathcal{Z}_\sigma^{\mathrm{subpre}}} : \mathcal{Z}_\sigma^{\mathrm{subpre}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{preproto}}$$

and

$$(\Pi_\sigma^{\mathrm{preproto}} \circ \Pi_\sigma^{\mathrm{prenuc}} \circ \widehat{\Pi}_\sigma^{\mathrm{precore}} \circ \Pi_\sigma^{\mathrm{subpre}})\big|_{\mathcal{Z}_\sigma^{\mathrm{subpre}}} : \mathcal{Z}_\sigma^{\mathrm{subpre}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{proto}}$$

are also diffeomorphisms.

Proof. Definition 6(1) gives

$$\Pi_\sigma^{\mathrm{subpre}}(\mathcal{B}_\sigma^{\mathrm{subpre}}) = \mathcal{B}_\sigma^{\mathrm{precore,tr}}$$

and

$$\Pi_\sigma^{\mathrm{subpre}} \circ \mathfrak{f}_\sigma^{\mathrm{subpre,tr}} = \mathrm{id}_{\mathcal{B}_\sigma^{\mathrm{precore,tr}}}.$$

Composing with the recorded transport-core data of Definition 5(1) yields

$$(\widehat{\Pi}_\sigma^{\mathrm{precore}} \circ \Pi_\sigma^{\mathrm{subpre}})(\mathcal{B}_\sigma^{\mathrm{subpre}}) = \widehat{\Pi}_\sigma^{\mathrm{precore}}(\mathcal{B}_\sigma^{\mathrm{precore,tr}}) = \mathcal{B}_\sigma^{\mathrm{prenuc}}$$

and

$$(\widehat{\Pi}_\sigma^{\mathrm{precore}} \circ \Pi_\sigma^{\mathrm{subpre}}) \circ \mathfrak{f}_\sigma^{\mathrm{subpre,tr}} \circ \mathfrak{f}_\sigma^{\mathrm{precore,tr}} = \widehat{\Pi}_\sigma^{\mathrm{precore}} \circ \mathfrak{f}_\sigma^{\mathrm{precore,tr}} = \mathrm{id}_{\mathcal{B}_\sigma^{\mathrm{prenuc}}}.$$

For shell transport, Definition 6(2) gives a diffeomorphism

$$\Pi_\sigma^{\mathrm{subpre}}\big|_{\mathcal{Z}_\sigma^{\mathrm{subpre}}} : \mathcal{Z}_\sigma^{\mathrm{subpre}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{precore,tr}},$$

while Definition 5(2) gives a diffeomorphism

$$\widehat{\Pi}_\sigma^{\mathrm{precore}}\big|_{\mathcal{Z}_\sigma^{\mathrm{precore,tr}}} : \mathcal{Z}_\sigma^{\mathrm{precore,tr}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{prenuc}}.$$

Their composition is therefore a diffeomorphism from $\mathcal{Z}_\sigma^{\mathrm{subpre}}$ to $\mathcal{Z}_\sigma^{\mathrm{prenuc}}$. Composing once more with the shell diffeomorphism of Definition 4(2) yields the displayed diffeomorphism to $\mathcal{Z}_\sigma^{\mathrm{preproto}}$, and composing finally with the shell diffeomorphism of Definition 3(2) yields the displayed diffeomorphism to $\mathcal{Z}_\sigma^{\mathrm{proto}}$. □

Proposition 3 (The benchmark package remains fixed)**.** *Every benchmark-faithful sub-pre-core package preserves the same benchmark one-metric package* (1)*. In particular, the operative metric remains exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged.*

Proof. This is part of benchmark faithfulness in Definition 6(3). The present note introduces only a deeper substrate-side source for the current transport core. It does not alter the benchmark exact-closure package. □

5. Main Theorem

Theorem 1 (Observer-relevant pre-core exact-shell transport core from exact-shell-transport-core–generating substrate sub-pre-core dynamics)**.** *Fix the primitive continuation data of Definition 1, the recorded proto-germ exact-shell target of Definition 2, the recorded current transport nucleus of Definition 3, the recorded observer-relevant pre-nucleus exact-shell core of Definition 4, the recorded*

observer-relevant pre-core exact-shell transport core of Definition 5, and a benchmark-faithful sub-pre-core package in the sense of Definition 6. Then, for each sector $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$, the current transport core is no longer an unexplained primitive stopping point on the active line: it is recovered canonically as the projected exact-shell transport image of the deeper sub-pre-core package. More precisely:

1. *the sub-pre-core package generates the recorded transport core by Proposition 1;*
2. *the exact-shell transport from the sub-pre-core package already carries the recorded downstream transport to the pre-nucleus exact-shell core, the current transport nucleus, and the fixed proto-germ exact-shell target by Proposition 2;*
3. *the benchmark boundary remains fixed by Proposition 3;*
4. *consequently the remaining honest burden below the previous transport-core frontier is no longer the observer-relevant pre-core exact-shell transport core $\mathfrak{S}^{\mathrm{tr}}$ as such. What remains is to derive or justify the deeper exact-shell-transport-core-generating substrate sub-pre-core dynamics $\mathfrak{U}^{\mathrm{subpre}}$ from still more primitive substrate structure, or else prove a still sharper collapse strictly inside that deeper package.*

Proof. Item (1) is Proposition 1. Item (2) is Proposition 2. Item (3) is Proposition 3. Item (4) is the routing consequence of the previous items together with the active safeguard against counting another same-output deeper-origin relay as new front-line progress when the live benchmark-relevant datum is unchanged. □

Corollary 1 (What must be done next)**.** *After Theorem 1, the next honest benchmark-facing manuscript must either:*

1. *derive or justify the exact-shell-transport-core-generating substrate sub-pre-core dynamics $\mathfrak{U}^{\mathrm{subpre}}$ from still more primitive substrate structure in a way that changes benchmark-facing status;*
2. *identify a genuinely smaller theorem-level observer-relevant datum strictly inside $\mathfrak{U}^{\mathrm{subpre}}$ and prove a sharper collapse to it;*
3. *do either of the previous two while preserving the same benchmark one-metric package and the same claim boundary.*

Another same-output deeper-origin relay that preserves the same transport core, another re-expansion back to the full pre-core package as if its screened ambient directions were still live, another reopening of the former pre-nucleus exact-shell-core frontier as if it were still the active burden, or any move that introduces a second operative metric, enlarges the ordinary matter law, or uses stronger promotion language does not satisfy the present burden. If a quantum continuation is pursued, it matters benchmark-facingly only insofar as it derives this same transport core, collapses the deeper sub-pre-core package further, or closes a benchmark derivation gate in a genuinely new way.

Proof. Theorem 1 shows that the current transport core is now a recovered target rather than the primitive stopping point of the active line. Therefore a subsequent manuscript must improve on that status rather than merely accompany it. What remains open is either to derive or justify the deeper sub-pre-core package itself from more primitive structure, or to discover a smaller theorem-level observer-relevant datum inside it. □

6. Conclusion

The positive result of this note is a disciplined deeper derivation step below the current transport-core frontier. The previous active-primary theorem already removed the full pre-core package as the live burden by collapsing it to a smaller observer-relevant exact-shell transport core. The present theorem then takes the next honest route available there: it shows that even that current transport core is not the primitive place where the active line has to stop, because it is recovered canonically from deeper exact-shell-transport-core-generating substrate sub-pre-core dynamics.

The benchmark boundary remains unchanged. The one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged. This remains a disciplined derivational gain inside the active program of Æther-flow Interpretation of Relativity, not a license for stronger promotion language.

That is the honest next burden after the present theorem. A new primary manuscript must derive or justify the deeper sub-pre-core dynamics from still more primitive substrate structure, or else prove a still sharper collapse strictly inside that package, while preserving the same benchmark one-metric package and the same claim boundary.